

#In this code we will demonstrate proposition 7.22. We will verify that there is a single configuration along the family described by γ_3 with zero total mass.

#Define the mass component ni , nj and nk

$$ni := \frac{y}{8} \cdot (s^{-3} - T) \cdot (z^{-3} - w^{-3}) :$$

$$nj := \frac{x}{8} \cdot (s^{-3} - T) \cdot (z^{-3} + w^{-3} - 2 \cdot x^{-3}) :$$

$$nk := \frac{x}{4} \cdot ((w^{-3} - x^{-3}) \cdot (y^{-3} - z^{-3}) + (z^{-3} - x^{-3}) \cdot (y^{-3} - w^{-3})) :$$

#Define the total mass $m0$

$$m0 := 2 \, ni + 2 \, nj + nk$$

$$m0 := \frac{y \left(\frac{1}{s^3} - T \right) \left(\frac{1}{z^3} - \frac{1}{w^3} \right)}{4} + \frac{x \left(\frac{1}{s^3} - T \right) \left(\frac{1}{z^3} + \frac{1}{w^3} - \frac{2}{x^3} \right)}{4} \quad (1)$$

$$+ \frac{x \left(\left(\frac{1}{w^3} - \frac{1}{x^3} \right) \left(\frac{1}{y^3} - \frac{1}{z^3} \right) + \left(\frac{1}{z^3} - \frac{1}{x^3} \right) \left(\frac{1}{y^3} - \frac{1}{w^3} \right) \right)}{4}$$

#The configurations described by γ_3 satisfy`

$$s := \frac{y}{2}; T := \frac{z^{-3} + w^{-3}}{2}$$

$$s := \frac{y}{2}$$

$$T := \frac{1}{2 z^3} + \frac{1}{2 w^3} \quad (2)$$

#Rewriting $m0$

$$m0$$

$$\frac{y \left(\frac{8}{y^3} - \frac{1}{2 z^3} - \frac{1}{2 w^3} \right) \left(\frac{1}{z^3} - \frac{1}{w^3} \right)}{4} + \frac{x \left(\frac{8}{y^3} - \frac{1}{2 z^3} - \frac{1}{2 w^3} \right) \left(\frac{1}{z^3} + \frac{1}{w^3} - \frac{2}{x^3} \right)}{4} \quad (3)$$

$$+ \frac{x \left(\left(\frac{1}{w^3} - \frac{1}{x^3} \right) \left(\frac{1}{y^3} - \frac{1}{z^3} \right) + \left(\frac{1}{z^3} - \frac{1}{x^3} \right) \left(\frac{1}{y^3} - \frac{1}{w^3} \right) \right)}{4}$$

#Collecting the numerator and denominator of $m0$

$$\text{numer}(m0); \text{denom}(m0)$$

$$-x \left(w^6 x^3 y^3 - 18 w^6 x^3 z^3 + w^6 x^2 y^4 - 16 w^6 x^2 y z^3 - 4 w^6 y^3 z^3 + 36 w^6 z^6 + 6 w^3 x^3 y^3 z^3 \right)$$

$$-18 w^3 x^3 z^6 + 16 w^3 x^2 y z^6 - 4 w^3 y^3 z^6 + x^3 y^3 z^6 - x^2 y^4 z^6) \\ 8 y^3 z^6 w^6 x^3$$

(4)

#Applying rational parameterization

$$\text{parametrization} := \left\{ \begin{array}{l} w = a + \frac{1}{a}, \\ z = b + \frac{1}{b}, \\ y = a - \frac{1}{a} + b - \frac{1}{b}, \\ x = a - \frac{1}{a} - b + \frac{1}{b} \end{array} \right\} :$$

simplify((expand(subs(parametrization, numer(m0)))));

simplify((expand(subs(parametrization, denom(m0)))));

$$-\frac{1}{a^{13} b^{13}} (2 (a b + 1) (-b + a) ((-b^{24} - 4 b^{22} - 8 b^{20} + 328 b^{18} + 1853 b^{16} + 4912 b^{14} \\ + 7200 b^{12} + 5952 b^{10} + 2433 b^8 + 368 b^6 - 8 b^4 - 4 b^2 - 1) a^{12} + (10 b^{21} + 65 b^{19} \\ + 274 b^{17} + 70 b^{15} - 933 b^{13} - 753 b^{11} + 76 b^9 + 212 b^7 - 29 b^5 - 16 b^3) a^9 + (40 b^{20} \\ + 262 b^{18} + 928 b^{16} + 1710 b^{14} + 1767 b^{12} + 1210 b^{10} + 696 b^8 + 274 b^6 + 40 b^4) a^8 \\ + (23 b^{19} + 190 b^{17} + 258 b^{15} - 216 b^{13} - 266 b^{11} - 44 b^9 + 60 b^7 - 5 b^5) a^7 + (20 b^{18} \\ + 246 b^{16} + 540 b^{14} + 432 b^{12} + 70 b^{10} + 14 b^8 + 22 b^6) a^6 + (15 b^{17} + 75 b^{15} + 89 b^{13} \\ + 135 b^{11} + 53 b^9 + 17 b^7) a^5 + (30 b^{16} + 50 b^{14} + 8 b^{12} - 66 b^{10} - 28 b^8) a^4 + (24 b^{15} \\ + 59 b^{13} + 73 b^{11} + 20 b^9) a^3 + 2 b^2 (b^{20} + 45 b^{18} + 292 b^{16} + 887 b^{14} + 1700 b^{12} \\ + 2184 b^{10} + 1887 b^8 + 1003 b^6 + 307 b^4 + 45 b^2 + 1) a^{10} - b (b^{22} - 29 b^{20} - 95 b^{18} \\ + 304 b^{16} + 733 b^{14} - 192 b^{12} - 992 b^{10} - 585 b^8 - 36 b^6 + 77 b^4 + 47 b^2 - 1) a^{13} + b (b^{22} \\ + 43 b^{20} + 141 b^{18} + 26 b^{16} - 581 b^{14} - 992 b^{12} - 192 b^{10} + 477 b^8 + 206 b^6 - 123 b^4 \\ - 61 b^2 - 1) a^{11} + 2 b^2 (b^{20} - 37 b^{18} - 27 b^{16} + 613 b^{14} + 2163 b^{12} + 3184 b^{10} + 2350 b^8 \\ + 729 b^6 - 12 b^4 - 37 b^2 + 1) a^{14} + (-2 b^{14} + 4 b^{12} - 2 b^{10}) a^2 + (-b^{13} + b^{11}) a + (b^{13} \\ - b^{11}) a^{23} + (-2 b^{14} + 4 b^{12} - 2 b^{10}) a^{22} + (-18 b^{15} - 41 b^{13} - 55 b^{11} - 14 b^9) a^{21} \\ + (30 b^{16} + 50 b^{14} + 8 b^{12} - 66 b^{10} - 28 b^8) a^{20} + (-33 b^{17} - 111 b^{15} - 107 b^{13} - 153 b^{11} \\ - 89 b^9 - 35 b^7) a^{19} + (14 b^{18} + 210 b^{16} + 450 b^{14} + 312 b^{12} - 20 b^{10} - 22 b^8 + 16 b^6) a^{18} \\ + (21 b^{19} - 60 b^{17} - 60 b^{15} + 364 b^{13} + 154 b^{11} - 226 b^9 - 190 b^7 - 3 b^5) a^{17} + (-42 b^{20} \\ - 66 b^{18} + 600 b^{16} + 2038 b^{14} + 2587 b^{12} + 1538 b^{10} + 368 b^8 - 54 b^6 - 42 b^4) a^{16} \\ + (14 b^{21} + 87 b^{19} - 108 b^{17} - 76 b^{15} + 1009 b^{13} + 929 b^{11} - 70 b^9 - 306 b^7 - 51 b^5 \\ - 20 b^3) a^{15} + a^{24} b^{12} + b^{12}))$$

$$\frac{8 (b^2 + 1)^6 (a^2 + 1)^6 (ab - 1)^3 (ab + 1)^3 (-b + a)^3 (a + b)^3}{a^{12} b^{12}} \quad (5)$$

#Define m0

$$\begin{aligned} m0 := & (a^{24} b^{12} + b^{12} + (b^{13} - b^{11}) a^{23} + (-2 b^{14} + 4 b^{12} - 2 b^{10}) a^{22} + (-18 b^{15} - 41 b^{13} \\ & - 55 b^{11} - 14 b^9) a^{21} + (30 b^{16} + 50 b^{14} + 8 b^{12} - 66 b^{10} - 28 b^8) a^{20} + (-33 b^{17} - 111 b^{15} \\ & - 107 b^{13} - 153 b^{11} - 89 b^9 - 35 b^7) a^{19} + (14 b^{18} + 210 b^{16} + 450 b^{14} + 312 b^{12} - 20 b^{10} \\ & - 22 b^8 + 16 b^6) a^{18} + (21 b^{19} - 60 b^{17} - 60 b^{15} + 364 b^{13} + 154 b^{11} - 226 b^9 - 190 b^7 \\ & - 3 b^5) a^{17} + (-42 b^{20} - 66 b^{18} + 600 b^{16} + 2038 b^{14} + 2587 b^{12} + 1538 b^{10} + 368 b^8 - 54 b^6 \\ & - 42 b^4) a^{16} + (14 b^{21} + 87 b^{19} - 108 b^{17} - 76 b^{15} + 1009 b^{13} + 929 b^{11} - 70 b^9 - 306 b^7 \\ & - 51 b^5 - 20 b^3) a^{15} + (-b^{24} - 4 b^{22} - 8 b^{20} + 328 b^{18} + 1853 b^{16} + 4912 b^{14} + 7200 b^{12} \\ & + 5952 b^{10} + 2433 b^8 + 368 b^6 - 8 b^4 - 4 b^2 - 1) a^{12} + (10 b^{21} + 65 b^{19} + 274 b^{17} + 70 b^{15} \\ & - 933 b^{13} - 753 b^{11} + 76 b^9 + 212 b^7 - 29 b^5 - 16 b^3) a^9 + (40 b^{20} + 262 b^{18} + 928 b^{16} \\ & + 1710 b^{14} + 1767 b^{12} + 1210 b^{10} + 696 b^8 + 274 b^6 + 40 b^4) a^8 + (23 b^{19} + 190 b^{17} + 258 b^{15} \\ & - 216 b^{13} - 266 b^{11} - 44 b^9 + 60 b^7 - 5 b^5) a^7 + (20 b^{18} + 246 b^{16} + 540 b^{14} + 432 b^{12} \\ & + 70 b^{10} + 14 b^8 + 22 b^6) a^6 + (15 b^{17} + 75 b^{15} + 89 b^{13} + 135 b^{11} + 53 b^9 + 17 b^7) a^5 \\ & + (30 b^{16} + 50 b^{14} + 8 b^{12} - 66 b^{10} - 28 b^8) a^4 + (24 b^{15} + 59 b^{13} + 73 b^{11} + 20 b^9) a^3 + (-2 b^{14} + 4 b^{12} - 2 b^{10}) a^2 \\ & + (-b^{13} + b^{11}) a + 2 b^2 (b^{20} + 45 b^{18} + 292 b^{16} + 887 b^{14} \\ & + 1700 b^{12} + 2184 b^{10} + 1887 b^8 + 1003 b^6 + 307 b^4 + 45 b^2 + 1) a^{10} + b (b^{22} + 43 b^{20} \\ & + 141 b^{18} + 26 b^{16} - 581 b^{14} - 992 b^{12} - 192 b^{10} + 477 b^8 + 206 b^6 - 123 b^4 - 61 b^2 - 1) a^{11} \\ & + 2 b^2 (b^{20} - 37 b^{18} - 27 b^{16} + 613 b^{14} + 2163 b^{12} + 3184 b^{10} + 2350 b^8 + 729 b^6 - 12 b^4 \\ & - 37 b^2 + 1) a^{14} - b (b^{22} - 29 b^{20} - 95 b^{18} + 304 b^{16} + 733 b^{14} - 192 b^{12} - 992 b^{10} - 585 b^8 \\ & - 36 b^6 + 77 b^4 + 47 b^2 - 1) a^{13} : \end{aligned}$$

#Define r3

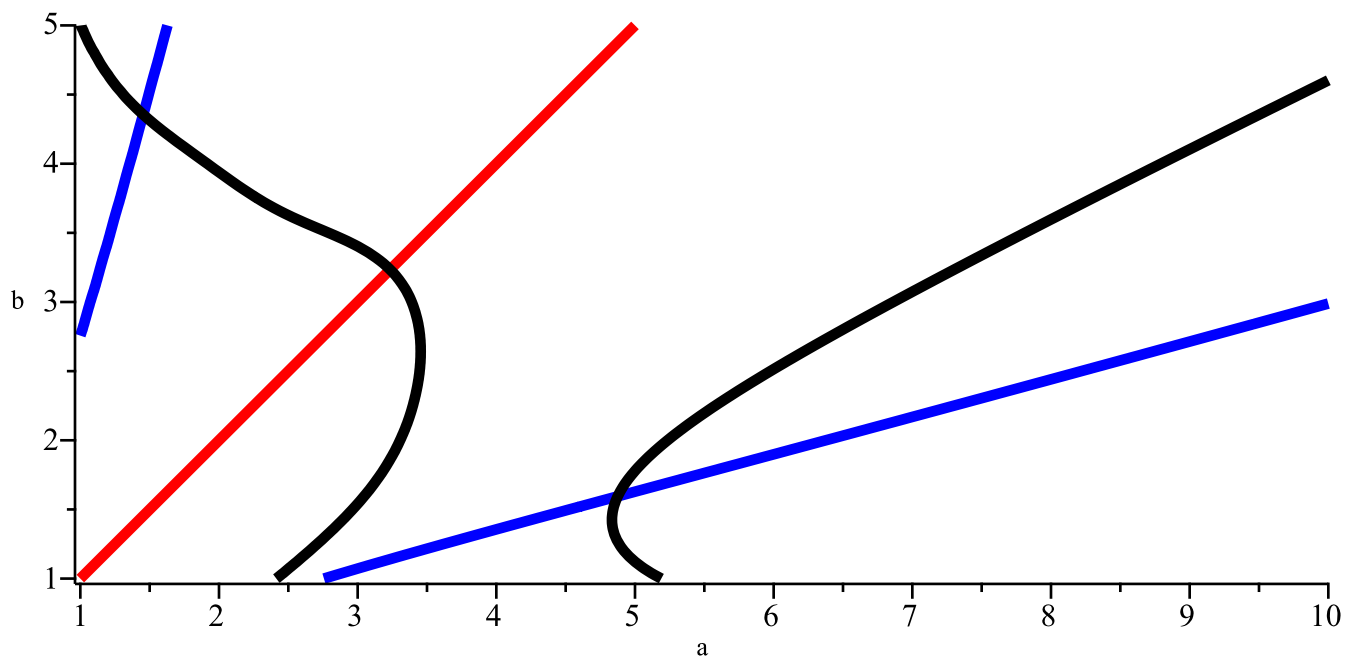
$$\begin{aligned} r3 := & (a^{22} b^{10} + (-6 b^{11} + 8 b^9) a^{21} + (15 b^{12} + 6 b^{10} + 30 b^8) a^{20} + (-18 b^{13} - 102 b^{11} \\ & + 162 b^9 + 74 b^7) a^{19} + (3 b^{14} + 198 b^{12} + 93 b^{10} + 600 b^8 + 145 b^6) a^{18} + (24 b^{15} - 96 b^{13} \\ & - 762 b^{11} + 1884 b^9 + 1380 b^7 + 252 b^5) a^{17} + (-294 b^{16} - 1740 b^{14} - 3162 b^{12} - 2920 b^{10} \\ & + 1908 b^8 + 1092 b^6 + 145 b^4) a^{16} + (24 b^{17} - 132 b^{15} - 2472 b^{13} - 8010 b^{11} + 3340 b^9 \\ & + 4692 b^7 + 1380 b^5 + 74 b^3) a^{15} + 3 b^2 (b^{16} - 580 b^{14} - 3862 b^{12} - 8784 b^{10} - 8307 b^8 \\ & - 1172 b^6 + 636 b^4 + 200 b^2 + 10) a^{14} + (-18 b^{19} - 96 b^{17} - 2472 b^{15} - 14520 b^{13} \\ & - 34416 b^{11} - 8268 b^9 + 3340 b^7 + 1884 b^5 + 162 b^3 + 8 b) a^{13} + (15 b^{20} + 198 b^{18} - 3162 b^{16} \\ & - 26352 b^{14} - 67416 b^{12} - 68232 b^{10} - 24921 b^8 - 2920 b^6 + 93 b^4 + 6 b^2 + 1) a^{12} - 6 b (b^{20} \\ & + 17 b^{18} + 127 b^{16} + 1335 b^{14} + 5736 b^{12} + 11692 b^{10} + 5736 b^8 + 1335 b^6 + 127 b^4 + 17 b^2 \\ & + 1) a^{11} + b^2 (b^{20} + 6 b^{18} + 93 b^{16} - 2920 b^{14} - 24921 b^{12} - 68232 b^{10} - 67416 b^8 - 26352 b^6 \\ & - 3162 b^4 + 198 b^2 + 15) a^{10} + (8 b^{21} + 162 b^{19} + 1884 b^{17} + 3340 b^{15} - 8268 b^{13} - 34416 b^{11} \\ & - 14520 b^9 - 2472 b^7 - 96 b^5 - 18 b^3) a^9 + (30 b^{20} + 600 b^{18} + 1908 b^{16} - 3516 b^{14} \\ & - 24921 b^{12} - 26352 b^{10} - 11586 b^8 - 1740 b^6 + 3 b^4) a^8 + (74 b^{19} + 1380 b^{17} + 4692 b^{15} \\ & + 3340 b^{13} - 8010 b^{11} - 2472 b^9 - 132 b^7 + 24 b^5) a^7 + (145 b^{18} + 1092 b^{16} + 1908 b^{14} \\ & - 2920 b^{12} - 3162 b^{10} - 1740 b^8 - 294 b^6) a^6 + (252 b^{17} + 1380 b^{15} + 1884 b^{13} - 762 b^{11} \\ & - 96 b^9 + 24 b^7) a^5 + (145 b^{16} + 600 b^{14} + 93 b^{12} + 198 b^{10} + 3 b^8) a^4 + (74 b^{15} + 162 b^{13} \\ & - 102 b^{11} - 18 b^9) a^3 + (30 b^{14} + 6 b^{12} + 15 b^{10}) a^2 + (8 b^{13} - 6 b^{11}) a + b^{12} : \end{aligned}$$

$r := a - b :$

with(plots) :

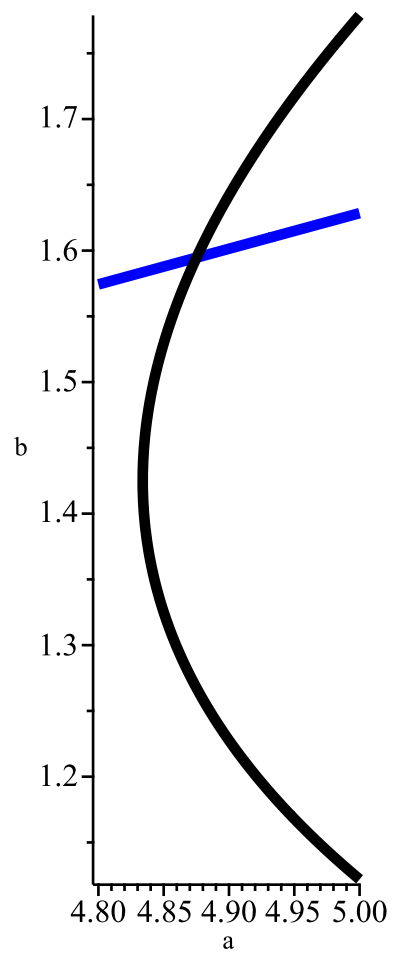
```
implicitplot([r3=0, r=0, m0=0], a=1 .. 10, b=1 .. (5),
  scaling=constrained,
  gridrefine=2,
  thickness=4,
  color=[blue, red, black, black],
  labels=["a", "b"],
  title="Graphic for r3=0 and m0=0");
```

Graphic for $r_3=0$ and $m_0=0$



```
implicitplot([r3=0, r2=0, m0=0], a=4.8 .. 5, b=1 .. (2),
  scaling=constrained,
  gridrefine=2,
  thickness=4,
  color=[blue, red, black],
  labels=["a", "b"],
  title="Graphic for  $r_3=0$  and  $m_0=0$ ");
```

Graphic for $r_3=0$ and $m_0=0$



#Approximate value for $\gamma_3(2)$

$fsolve(eval(r3, b=2)=0, a, 4 \dots \infty)$

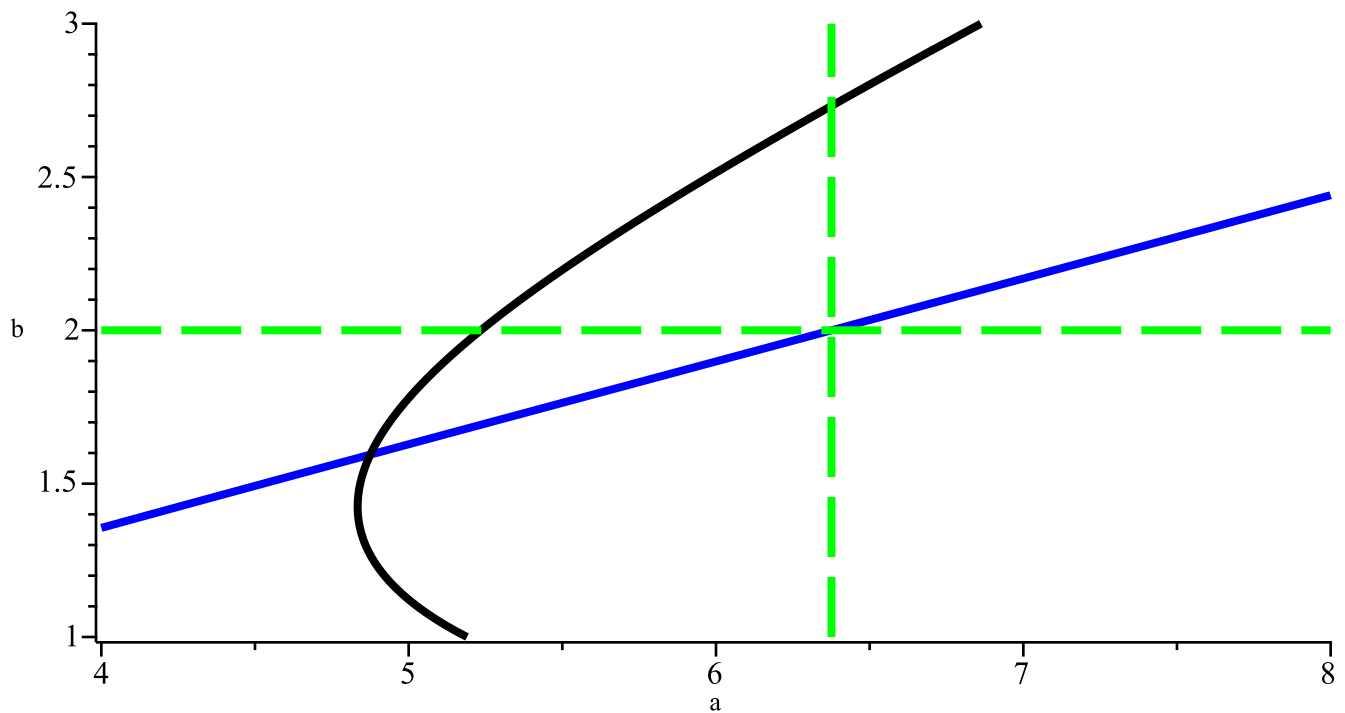
6.375758695

(6)

with(plots) :

```
implicitplot([r3=0, r2=0, m0=0, 2 - b=0, 6.375758695 - a=0], a=4 .. 8, b=1 .. (3),
  scaling=constrained,
  gridrefine=2,
  thickness=3,
  color=[blue, red, black, green, green],
  linestyle=[1, 1, 1, 3, 3],
  labels=["a", "b"],
  title="Graphic for r3=0 and m0=0");
```

Graphic for $r_3=0$ and $m_0=0$



#Locates a range for $\gamma_3(2)$

sturm(sturmseq(eval(r3, b = 2), a), a), a, 6, 7)

1

(7)

#6 < $\gamma_3(2)$ < 7

#Calculate $m_0(a, 2)$ and its derivative

eval(m0, b = 2);

diff(eval(m0, b = 2), a)

$4096 a^{24} + 6144 a^{23} - 18432 a^{22} - 1045504 a^{21} + 2743296 a^{20} - 9202560 a^{19} + 26058240 a^{18}$
 $+ 4336800 a^{17} + 23631840 a^{16} + 68418560 a^{15} + 99129192 a^{14} + 42277098 a^{13}$
 $+ 282188911 a^{12} + 148606902 a^{11} + 450129192 a^{10} + 84137440 a^9 + 208131840 a^8$
 $+ 43087200 a^7 + 32058240 a^6 + 5458560 a^5 + 2743296 a^4 + 1429504 a^3 - 18432 a^2$
 $- 6144 a + 4096$

$98304 a^{23} + 141312 a^{22} - 405504 a^{21} - 21955584 a^{20} + 54865920 a^{19} - 174848640 a^{18}$
 $+ 469048320 a^{17} + 73725600 a^{16} + 378109440 a^{15} + 1026278400 a^{14} + 1387808688 a^{13}$
 $+ 549602274 a^{12} + 3386266932 a^{11} + 1634675922 a^{10} + 4501291920 a^9 + 757236960 a^8$
 $+ 1665054720 a^7 + 301610400 a^6 + 192349440 a^5 + 27292800 a^4 + 10973184 a^3$
 $+ 4288512 a^2 - 36864 a - 6144$

(8)

#Verify that the derivative of $m_0(a, 2)$ does not vanish in the interval [6, 7]

sturm(sturmseq(diff(eval(m0, b = 2), a), a), a), a, 6, 7)

0

(9)

#Verify that $m_0(6, 2)$ and $m_0(7, 2)$ are positive

eval(eval(m0, b = 2), a = 6) ; eval(eval(m0, b = 2), a = 7)

6149282430277870428160

455465639195316474609375

(10)

#Therefore, $m_0(\gamma_3(2), 2)$ is positive and $m_0(\gamma_3(2), 2)$ is negative.

#This concludes the existence of a point $(\gamma_3(b), b)$ with $m_0 = 0$


```
#Compute the resultant R1
```

$$bfator := [op(factor(R1))]:$$

$$nops(bfator);$$

$$bfator[1] = -28991029248 \quad (12)$$

$$bfator[4] \quad (b^2 + 1)^{162} \quad (14)$$

$$degree(bfator[3], b)$$

204
(15)

$$seq\left(sign\left(evalf\left(coeff\left(numer\left(eval(bfactor[3], b = \frac{1+2d}{1+d})\right), d, i\right)\right), i = 204 \dots 0, -1\right);$$

[illegible]

$$seq\left(sign\left(evalf\left(coeff\left(numer\left(eval\left(bfactor[3], b = \frac{2 + 4d}{1 + d}\right)\right), d, i\right)\right), i = 204 .. 0, -1\right);$$

[illegible]

#Count the number of zeros in bfator[3] in the interval [4,5]

#The factor $bfator[3]$ has at most three roots between 1 and 5

$$fsolve(bfator[3]=0, b, 1..5);$$

$$1.594639279, 1.702227587, 4.355042597 \quad (19)$$
[illegible]

[illegible]

#Count the number of zeros in afator[3] in the interval $[3, 4]$

#The factor $afator[3]$ has at most three roots between 1 and 5

$$\begin{aligned} &fsolve(afator[3]=0, a, 1 \dots 5); \\ &1.216407333, 1.453232550, 1.530050281, 2.179476594, 4.875116002 \end{aligned} \quad (29)$$
[illegible]

#Thus, the candidate points (a,b) for the intersection of $r_3=0$ and $m_0=0$ are of the form $a \in A$ and $b \in B$ where

$A := [1.216407333, 1.453232550, 1.530050281, 2.179476594, 4.875116002] :$

$B := [1.594639279, 1.702227587, 4.355042597] :$

$Points := [seq(seq([a, b], b \in B), a \in A)] :$

$Points_filtered := select(p \rightarrow p[2] < p[1], Points);$

$Points_filtered := [[2.179476594, 1.594639279], [2.179476594, 1.702227587], [4.875116002, 1.594639279], [4.875116002, 1.702227587], [4.875116002, 4.355042597]]$ (31)

with(plots) :

with(plottools) :

```
p1 := implicitplot(
  [r3 = 0, r = 0],
  a = 2 .. 5, b = 1 .. 4.5,
  scaling = constrained,
  gridrefine = 2,
  thickness = 2,
  color = [blue, red],
  labels = ["a", "b"],
  title = "Graphic for r1=0 in A1"
) :
```

rectangles

```
R1 := rectangle([2, 1.5], [3, 2],
  color = silver, transparency = 0.5,
  linestyle = 3, # pontilhado
  thickness = 2) :
R2 := rectangle([4, 1.65], [5, 1.8],
  color = silver, transparency = 0.5,
  linestyle = 3, # pontilhado
  thickness = 2) :
R3 := rectangle([4, 4], [5, 4.5],
  color = silver, transparency = 0.5,
  linestyle = 3, # pontilhado
  thickness = 2) :
```

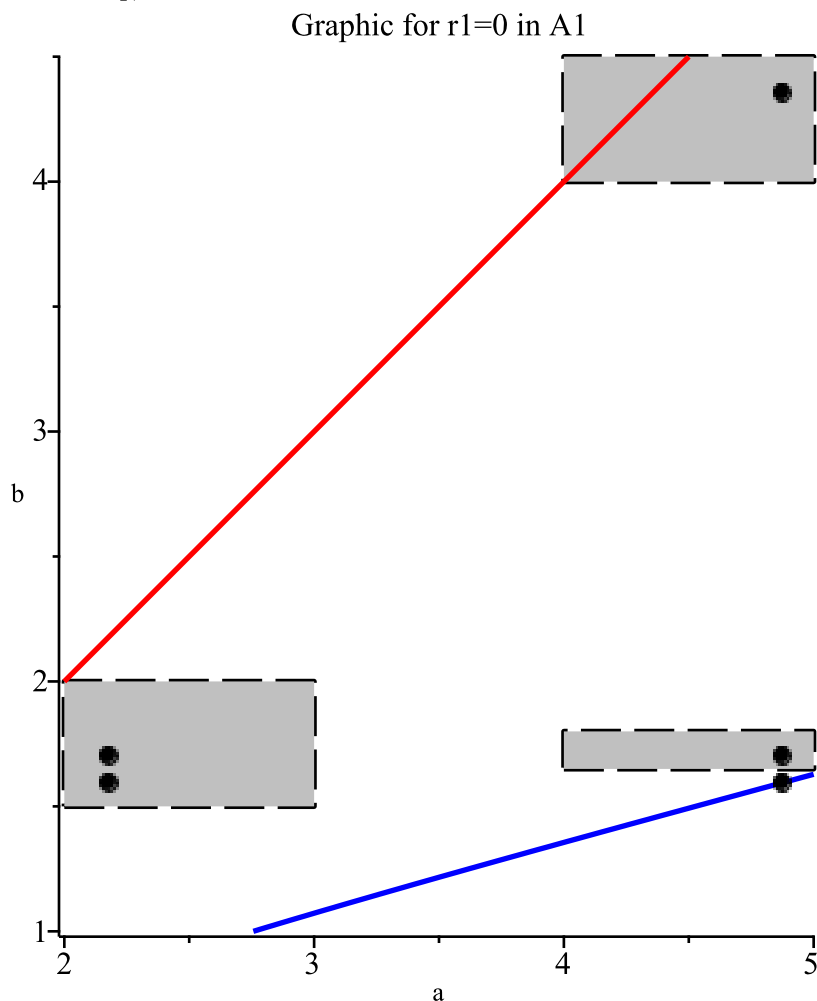
points

```
P := pointplot(
  Points_filtered,
  symbol = solidcircle,
  symbolsize = 12,
  color = black
```

) :

sobrepor tudo

display([p1, P, R1, R2, R3]);



#By covering the previous points with rectangles, we will show that the curve $r3=0$ does not intersect them.

restart;

#Define r3

$$\begin{aligned}
 r3 := & (a^{22} b^{10} + (-6 b^{11} + 8 b^9) a^{21} + (15 b^{12} + 6 b^{10} + 30 b^8) a^{20} + (-18 b^{13} - 102 b^{11} + 162 b^9 \\
 & + 74 b^7) a^{19} + (3 b^{14} + 198 b^{12} + 93 b^{10} + 600 b^8 + 145 b^6) a^{18} + (24 b^{15} - 96 b^{13} - 762 b^{11} \\
 & + 1884 b^9 + 1380 b^7 + 252 b^5) a^{17} + (-294 b^{16} - 1740 b^{14} - 3162 b^{12} - 2920 b^{10} + 1908 b^8 \\
 & + 1092 b^6 + 145 b^4) a^{16} + (24 b^{17} - 132 b^{15} - 2472 b^{13} - 8010 b^{11} + 3340 b^9 + 4692 b^7 \\
 & + 1380 b^5 + 74 b^3) a^{15} + 3 b^2 (b^{16} - 580 b^{14} - 3862 b^{12} - 8784 b^{10} - 8307 b^8 - 1172 b^6 \\
 & + 636 b^4 + 200 b^2 + 10) a^{14} + (-18 b^{19} - 96 b^{17} - 2472 b^{15} - 14520 b^{13} - 34416 b^{11} \\
 & - 8268 b^9 + 3340 b^7 + 1884 b^5 + 162 b^3 + 8 b) a^{13} + (15 b^{20} + 198 b^{18} - 3162 b^{16} - 26352 b^{14} \\
 & - 67416 b^{12} - 68232 b^{10} - 24921 b^8 - 2920 b^6 + 93 b^4 + 6 b^2 + 1) a^{12} - 6 b (b^{20} + 17 b^{18}
 \end{aligned}$$

[illegible]

[illegible]

restart;

#Define r3

$$\begin{aligned}
 r3 := & (a^{22} b^{10} + (-6 b^{11} + 8 b^9) a^{21} + (15 b^{12} + 6 b^{10} + 30 b^8) a^{20} + (-18 b^{13} - 102 b^{11} + 162 b^9 \\
 & + 74 b^7) a^{19} + (3 b^{14} + 198 b^{12} + 93 b^{10} + 600 b^8 + 145 b^6) a^{18} + (24 b^{15} - 96 b^{13} - 762 b^{11} \\
 & + 1884 b^9 + 1380 b^7 + 252 b^5) a^{17} + (-294 b^{16} - 1740 b^{14} - 3162 b^{12} - 2920 b^{10} + 1908 b^8 \\
 & + 1092 b^6 + 145 b^4) a^{16} + (24 b^{17} - 132 b^{15} - 2472 b^{13} - 8010 b^{11} + 3340 b^9 + 4692 b^7 \\
 & + 1380 b^5 + 74 b^3) a^{15} + 3 b^2 (b^{16} - 580 b^{14} - 3862 b^{12} - 8784 b^{10} - 8307 b^8 - 1172 b^6 \\
 & + 636 b^4 + 200 b^2 + 10) a^{14} + (-18 b^{19} - 96 b^{17} - 2472 b^{15} - 14520 b^{13} - 34416 b^{11} \\
 & - 8268 b^9 + 3340 b^7 + 1884 b^5 + 162 b^3 + 8 b) a^{13} + (15 b^{20} + 198 b^{18} - 3162 b^{16} - 26352 b^{14} \\
 & - 67416 b^{12} - 68232 b^{10} - 24921 b^8 - 2920 b^6 + 93 b^4 + 6 b^2 + 1) a^{12} - 6 b (b^{20} + 17 b^{18} \\
 & + 127 b^{16} + 1335 b^{14} + 5736 b^{12} + 11692 b^{10} + 5736 b^8 + 1335 b^6 + 127 b^4 + 17 b^2 + 1) a^{11} \\
 & + b^2 (b^{20} + 6 b^{18} + 93 b^{16} - 2920 b^{14} - 24921 b^{12} - 68232 b^{10} - 67416 b^8 - 26352 b^6 \\
 & - 3162 b^4 + 198 b^2 + 15) a^{10} + (8 b^{21} + 162 b^{19} + 1884 b^{17} + 3340 b^{15} - 8268 b^{13} - 34416 b^{11} \\
 & - 14520 b^9 - 2472 b^7 - 96 b^5 - 18 b^3) a^9 + (30 b^{20} + 600 b^{18} + 1908 b^{16} - 3516 b^{14} \\
 & - 24921 b^{12} - 26352 b^{10} - 11586 b^8 - 1740 b^6 + 3 b^4) a^8 + (74 b^{19} + 1380 b^{17} + 4692 b^{15} \\
 & + 3340 b^{13} - 8010 b^{11} - 2472 b^9 - 132 b^7 + 24 b^5) a^7 + (145 b^{18} + 1092 b^{16} + 1908 b^{14} \\
 & - 2920 b^{12} - 3162 b^{10} - 1740 b^8 - 294 b^6) a^6 + (252 b^{17} + 1380 b^{15} + 1884 b^{13} - 762 b^{11} \\
 & - 96 b^9 + 24 b^7) a^5 + (145 b^{16} + 600 b^{14} + 93 b^{12} + 198 b^{10} + 3 b^8) a^4 + (74 b^{15} + 162 b^{13} \\
 & - 102 b^{11} - 18 b^9) a^3 + (30 b^{14} + 6 b^{12} + 15 b^{10}) a^2 + (8 b^{13} - 6 b^{11}) a + b^{12} :
 \end{aligned}$$

#Apply the Möbius transformation on the rectangle $(4, 5) \times (4, 4.5)$ that contains one point,
constructing r3

$$\text{moebius} := \left\{ \begin{array}{l} a = \frac{4 + 5 \cdot c}{1 + c}, \\ b = \frac{4 + \frac{9}{5} \cdot d}{1 + d} \end{array} \right\} :$$

r3 := collect(numer(expand(subs(moebius, r3))), c) :

#Compute the degree of r3 in c

degree(r3, c);

22

(35)

#Define the coefficients ci = coeff(r3, c, i)

for i from 0 to 32 do

 c[i] := coeff(r3, c, i);

od:

```
#Compute the all degree of ci in d
seq(degree(c[i], d), i=0..22);
```

[illegible]

(36)

#From here we will analyze the signal variation of all coefficients c_i

$$seq(seq(sign(coeff(c[j], d, i)), i = 22..0, -1), j = 22..0, -1);$$
[illegible][illegible]

restart;

#Define r3

$$\begin{aligned}
 r3 := & (a^{22} b^{10} + (-6 b^{11} + 8 b^9) a^{21} + (15 b^{12} + 6 b^{10} + 30 b^8) a^{20} + (-18 b^{13} - 102 b^{11} + 162 b^9 \\
 & + 74 b^7) a^{19} + (3 b^{14} + 198 b^{12} + 93 b^{10} + 600 b^8 + 145 b^6) a^{18} + (24 b^{15} - 96 b^{13} - 762 b^{11} \\
 & + 1884 b^9 + 1380 b^7 + 252 b^5) a^{17} + (-294 b^{16} - 1740 b^{14} - 3162 b^{12} - 2920 b^{10} + 1908 b^8 \\
 & + 1092 b^6 + 145 b^4) a^{16} + (24 b^{17} - 132 b^{15} - 2472 b^{13} - 8010 b^{11} + 3340 b^9 + 4692 b^7 \\
 & + 1380 b^5 + 74 b^3) a^{15} + 3 b^2 (b^{16} - 580 b^{14} - 3862 b^{12} - 8784 b^{10} - 8307 b^8 - 1172 b^6 \\
 & + 636 b^4 + 200 b^2 + 10) a^{14} + (-18 b^{19} - 96 b^{17} - 2472 b^{15} - 14520 b^{13} - 34416 b^{11} \\
 & - 8268 b^9 + 3340 b^7 + 1884 b^5 + 162 b^3 + 8 b) a^{13} + (15 b^{20} + 198 b^{18} - 3162 b^{16} - 26352 b^{14} \\
 & - 67416 b^{12} - 68232 b^{10} - 24921 b^8 - 2920 b^6 + 93 b^4 + 6 b^2 + 1) a^{12} - 6 b (b^{20} + 17 b^{18} \\
 & + 127 b^{16} + 1335 b^{14} + 5736 b^{12} + 11692 b^{10} + 5736 b^8 + 1335 b^6 + 127 b^4 + 17 b^2 + 1) a^{11} \\
 & + b^2 (b^{20} + 6 b^{18} + 93 b^{16} - 2920 b^{14} - 24921 b^{12} - 68232 b^{10} - 67416 b^8 - 26352 b^6 \\
 & - 3162 b^4 + 198 b^2 + 15) a^{10} + (8 b^{21} + 162 b^{19} + 1884 b^{17} + 3340 b^{15} - 8268 b^{13} - 34416 b^{11} \\
 & - 14520 b^9 - 2472 b^7 - 96 b^5 - 18 b^3) a^9 + (30 b^{20} + 600 b^{18} + 1908 b^{16} - 3516 b^{14} \\
 & - 24921 b^{12} - 26352 b^{10} - 11586 b^8 - 1740 b^6 + 3 b^4) a^8 + (74 b^{19} + 1380 b^{17} + 4692 b^{15} \\
 & + 3340 b^{13} - 8010 b^{11} - 2472 b^9 - 132 b^7 + 24 b^5) a^7 + (145 b^{18} + 1092 b^{16} + 1908 b^{14} \\
 & - 2920 b^{12} - 3162 b^{10} - 1740 b^8 - 294 b^6) a^6 + (252 b^{17} + 1380 b^{15} + 1884 b^{13} - 762 b^{11} \\
 & - 96 b^9 + 24 b^7) a^5 + (145 b^{16} + 600 b^{14} + 93 b^{12} + 198 b^{10} + 3 b^8) a^4 + (74 b^{15} + 162 b^{13} \\
 & - 102 b^{11} - 18 b^9) a^3 + (30 b^{14} + 6 b^{12} + 15 b^{10}) a^2 + (8 b^{13} - 6 b^{11}) a + b^{12} :
 \end{aligned}$$

#Apply the Möbius transformation on the rectangle $(4, 5) \times (1.65, 1.8)$ that contains one point,
constructing r3

$$\text{moebius} := \left\{ \begin{array}{l} a = \frac{4 + 5 \cdot c}{1 + c}, \\ b = \frac{\frac{33}{20} + \frac{9}{5} \cdot d}{1 + d} \end{array} \right\} :$$

r3 := collect(numer(expand(subs(moebius, r3))), c) :

#Compute the degree of r3 in c

degree(r3, c);

22

(38)

#Define the coefficients ci = coeff(r3, c, i)

for i from 0 to 32 **do**

 c[i] := coeff(r3, c, i);

od:

```
#Compute the all degree of ci in d
seq(degree(c[i], d), i = 0..22);
```

[illegible]

#From here we will analyze the signal variation of all coefficients c_i

[illegible]